

Homework 3 – Due February 10 at 11:59 PM

Please submit a pdf of completed HW through Canvas with all problems in a single file. For scanning handwritten work for submission, I've personally found the Genius Scan app works well. You must show your work to earn full credit. Late work will not be accepted.

1. Fill in the table below for R-134a: (Note this is not H₂O. R-134a is a refrigerant with its own tables in the textbook.)

T (°C)	P (kPa)	u (kJ/kg)	Phase
20		95	
-12			Saturated liquid
	400	300	
8	600		

2. Superheated water vapor initially at $P = 10$ bar and $T = 400$ °C is contained within a piston-cylinder assembly. The water is cooled at a constant volume until its temperature is 150 °C. The water is then condensed isothermally until it is a saturated liquid. Considering the water as the system,
 - a. Sketch the process on a T-v diagram.
 - b. Evaluate the specific work (work per unit mass) of the process. (*Answer check: $w = -145.4$ kJ/kg*)
3. Determine the compressibility factor for water vapor at 200 bar and 470 °C using the following methods:
 - a. Data from the compressibility charts. (*Answer check: $Z = 0.78$*)
 - b. Using the ideal gas law and data from Table A-4. (*Answer check: $Z = 0.790$*)
4. A tank contains 50 L of nitrogen (N₂) at -20 °C and 10 MPa. Determine the mass of nitrogen using the following methods:
 - a. The unmodified ideal gas law (without considering compressibility). (*Answer check: $m = 6.65$ kg*)
 - b. The ideal gas law and data from the compressibility chart. (*Answer check: $m = 6.93$ kg*)
5. A quantity of water is at $P = 15$ MPa and $T = 100$ °C. Evaluate the specific volume and specific enthalpy using two different methods:

- a. Data from Table A-5. (Answer check: $v = 1.0361 \times 10^{-3} \text{ m}^3/\text{kg}$, $h = 430.28 \text{ kJ/kg}$)
 - b. Saturated liquid data from Table A-2, along with the equation for approximating enthalpy of an approximately incompressible substance:
 $h(T, P) = h_f(T) + v_f(T) \cdot [P - P_{sat}(T)]$. (Answer check: $v = 1.0435 \times 10^{-3} \text{ m}^3/\text{kg}$, $h = 434.48 \text{ kJ/kg}$)
6. Water is contained in a piston-cylinder assembly (as shown in the figure below), initially at 1.5 bar and a quality of 20%. The water is heated at a constant pressure until the piston hits the stops. From there, heating continues until the water is saturated vapor.
- a. Sketch the processes of the water on a T-v diagram.
 - b. For the overall process of the water, evaluate the specific work and heat transfer per unit mass. Neglect changes in kinetic and potential energy. (This problem is tricky – the process is somewhat similar to the process in problem 2, but in reverse.) (Answer check: $w = 20.94 \text{ kJ/kg}$, $q = 1705 \text{ kJ/kg}$)

